

“Precision” is not one thing. It depends on what is being resampled.

Paired subsampling. Sample the *same* student indices at both time points: draw (U_i, V_i) pairs by resampling indices $\{i_1, \dots, i_n\}$ from the subgroup. This preserves the latent covariance $\text{Cov}(U_i, V_i)$ and typically yields *tighter* repeated-sampling precision.

Independent cross-sectional resampling. Sample grade- t_1 and grade- t_2 populations independently: different students contribute to F_U and F_V . This matches the NAEP/TIMSS use case and produces *wider* uncertainty because no within-student covariance anchors the bridge.

Without distinguishing these two worlds, readers will conflate:

- linked-subsample stability (optimistic),
 - unlinked-cross-sectional operating precision (realistic),
- and
- bridge accuracy (a separate axis).

Those are three different beasts. Panel C makes the design dependence visible so that the precision curves in Panel D can be read correctly.³